



R1.3 Energy from fuels

Reactivity 1 · What drives chemical reactions? · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Fuels release energy through exothermic combustion reactions, and the enthalpy of combustion (ΔH_c° , the energy released per mole of fuel burned completely in oxygen under standard conditions) is used to compare the energy content of different fuels; specific energy (energy per unit mass, kJ g^{-1}) is often more useful than molar enthalpy of combustion for comparing fuels of different molar mass. Fossil fuels (coal, oil, natural gas) are non-renewable, finite, and their combustion releases greenhouse gases such as CO_2 (contributing to climate change) and, with incomplete combustion or impurities, pollutants such as CO, unburned hydrocarbons, and SO_2/NO_x ; alternative energy sources include biofuels, hydrogen, and renewable sources such as solar/wind, which aim to reduce reliance on fossil fuels and lower carbon emissions.

Worked example: follow the reasoning

0.500 g of ethanol, $\text{C}_2\text{H}_5\text{OH}$ ($M = 46.0 \text{ g mol}^{-1}$), is burned completely and releases 14.9 kJ of heat energy. Calculate the specific energy (in kJ g^{-1}) and the molar enthalpy of combustion (in kJ mol^{-1}) of ethanol.

Step 1: Specific energy = heat released / mass = $14.9 / 0.500 = 29.8 \text{ kJ g}^{-1}$.

Step 2: $n(\text{ethanol}) = 0.500 / 46.0 = 0.01087 \text{ mol}$.

Step 3: $\Delta H_c = -\text{heat released} / n = -14.9 / 0.01087 = -1371 \text{ kJ mol}^{-1}$ (negative since combustion is exothermic).

Answer: Specific energy $\approx 29.8 \text{ kJ g}^{-1}$; $\Delta H_c(\text{ethanol}) \approx -1370 \text{ kJ mol}^{-1}$.

Quick check · multiple choice

Why is the specific energy (kJ g^{-1}) of hydrogen gas much greater than that of a typical hydrocarbon fuel such as octane?

A Hydrogen has a much larger molar mass than octane

B Hydrogen has a very small molar mass, so its energy per mole is spread over a much smaller mass

C Hydrogen releases more heat per mole than octane

D Hydrogen combustion is less exothermic than octane combustion

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A student compares two fuels, methane (CH_4 , $M = 16.0 \text{ g mol}^{-1}$, $\Delta H_c^\circ = -890 \text{ kJ mol}^{-1}$) and propane (C_3H_8 , $M = 44.0 \text{ g mol}^{-1}$, $\Delta H_c^\circ = -2220 \text{ kJ mol}^{-1}$). [5]
- (a) Calculate the specific energy (kJ g^{-1}) of each fuel. [2]
- (b) State, with reference to your answers in (a), which fuel releases more energy per gram. [1]
- (c) Write the balanced equation for the complete combustion of propane. [1]
- (d) State one environmental disadvantage of using these fuels compared with a fuel such as hydrogen. [1]

Final answer: _____



Answers and explanation

Question 1 [5]

- (a) Methane: $890/16.0 = 55.6 \text{ kJ g}^{-1}$. Propane: $2220/44.0 = 50.5 \text{ kJ g}^{-1}$. [2]
(b) Methane releases more energy per gram ($55.6 > 50.5 \text{ kJ g}^{-1}$). [1]
(c) $\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{l/g})$. [1]
(d) Complete combustion of hydrocarbon fuels such as methane and propane produces CO_2 , a greenhouse gas that contributes to global warming/climate change, whereas hydrogen combustion produces only water. [1]

Quick check answer

B. Specific energy = $\Delta H_c / \text{molar mass}$; hydrogen's very low molar mass (2.0 g mol^{-1}) means even a moderate molar enthalpy of combustion gives a very high energy release per gram.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Chemistry HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.